

Md Monirul Islam

+8801710096305 monirulislam.acad@gmail.com [Personal Website](#) [LinkedIn](#) [Google Scholar](#) [Researchgate](#)

Education

- **Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh
B.Sc. in Electrical and Electronic Engineering March 2018 - May 2023
CGPA:3.85/4.00 (**3.98/4.00 in the final year**)
Relevant Courses: Electrical & Electronic Circuits, Calculus I & II, Ordinary & Partial Differential Equations, Linear Algebra, Communication System, Digital Signal Processing, Digital Electronics, Power Electronics, Control Systems, Microprocessors and Embedded systems, Analog Integrated Circuits and Design, VLSI Circuits and Design

Research Interests

- Robotics • Autonomous Systems • Motion Control • Path Planning • Machine Learning • Wearable Robotics
- Mechatronic Design • Human-Robot Interaction • Robot Perception • Additive Manufacturing • Embedded System

Research Experience

- **Research on α -Graphyne Ballistic Nanoribbon FET:** Modeled ballistic transport in α -graphyne nanoribbons. Quantified I-V behavior and subthreshold slope to guide low-power device design. *Supervisor: Dr. Mahbub Alam*
- **Remotely Operated Vehicle (ROV) Development:** Studied control-architecture choices and latency budgets for a 6-DOF arm ROV. Derived tuning criteria and power-budget models from bench tests. *Supervisor: Rashed Mujib Noman*
- **Industrial-Grade 3D Printer Development:** Analyzed frame rigidity, thermal management, and motion-control limits for large build volumes. Produced controller and firmware parameter guidelines. *Supervisor: Rashed Mujib Noman*
- **Autonomous Target Tracking Drone:** Evaluated vision-in-the-loop guidance pipelines on embedded hardware. Compared trackers and closed-loop stability metrics for real-time pursuit. *Supervisor: Rashed Mujib Noman*
- **Man-Transportable ROV with 7-DOF Arm:** Designed ROS node graph and kinematic-control test plan. Focused on solver accuracy, rate scheduling, and safety states. *Supervisor: Rashed Mujib Noman*
- **Autonomous Guided Vehicle (AGV) for Factory Floor:** Prototyped fleet management, localization experiments and navigation strategies for structured indoor layouts. Measured path efficiency, throughput, and collision margins. *Supervisor: Rashed Mujib Noman*

Publications

- H. Khan, **Md Monirul Islam**, R.I. Roya, S.N. Azad, "Performance Analysis of an α -Graphyne Nano-Field Effect Transistor," Novel Electronics Devices Integrated with 2D Quantum Materials, Micromachines, MDPI. doi: [10.3390/mi14071385](https://doi.org/10.3390/mi14071385)
- Md Shakil Tanvir, Fiaz Ahmed Tonmoy, **Md Monirul Islam**, Mainul Islam, Mahamudul Hasan, Raihan Ur Rashid Razeen, "Optimizing Master-Slave Broadcast Communication in Multi-Node Networks Using RS485 Standard", 2024, 27th International Conference on Computer and Information Technology (ICCIT), IEEE. doi: [10.1109/ICCIT64611.2024.11022430](https://doi.org/10.1109/ICCIT64611.2024.11022430)
- Md Shakil Tanvir, **Md Monirul Islam**, Fiaz Ahmed Tonmoy, Mainul Islam, Mahamudul Hasan, Raihan Ur Rashid Razeen, "A Dual-Network Feedback Communication System for Multi-Node Unmanned Robotic Control", [Submitted](#)
- **Md Monirul Islam**, Fiaz Ahmed Tonmoy, Md Shakil Tanvir, Mainul Islam, Mahamudul Hasan, Raihan Ur Rashid Razeen, "Differential-Drive Synchronization Using Genetic Algorithm Optimized Cascaded PID Controllers on STM32 Based MCU", [Submitted](#)

Professional Experience

- **Research and Development Unit, Cybernetics Hi Tech Solutions (Pvt) Ltd** Dhaka, Bangladesh
R&D Engineer and Team Lead(June 2023 - present)
 - Developed and field-tested a motion control algorithm for a Remotely Operated Vehicle (ROV) with a 6-degree-of-freedom robotic arm, enhancing precision, stability, and functionality.
 - Engineered and optimized a speed synchronization algorithm for BLDC motors, significantly improving performance and ensuring accurate motor coordination across multiple drive units.
 - Designed and implemented a robust electrical communication system to coordinate motion control and sensor feedback across each DOF of the ROV.
 - Designed and fabricated essential PCBs for control and communication subsystems using CAD tools and CNC manufacturing, ensuring high reliability and compact form factors.
 - Developed a power distribution and management system for the ROV, enabling seamless and efficient energy flow during extended field operations.

- Designed and manufactured a large-volume, industrial-grade 3D printer, including thermal control, motion systems, and firmware integration.
- Participated in an industrial automation project for real-time pet food sorting using sensor arrays and actuated mechanisms, demonstrating flexibility across diverse automation challenges.
- Designed and optimized a heavy-lift cargo drone (30–100 kg capacity), focusing on advanced battery management for extended endurance, flight stability, structural integrity, and propulsion systems.
- Designed and implemented a real-time object tracking drone by integrating an onboard flight computer with camera-based vision, enabling autonomous target detection and dynamic flight control for reliable pursuit.
- Working on integrating ROS (Robot Operating System) for the control of an advanced man-transportable ROV with a 7-DOF kinematically controlled arm, supporting modular software development and real-time control.
- Contributing to the development of a fleet management, autonomous navigation and localization system for a factory floor Autonomous Guided Vehicle (AGV), aimed at optimizing industrial logistics and reducing manual handling.

- **Ulkasemi Pvt. Limited** Dhaka, Bangladesh
Intern(Dec 2022 - Mar 2023)
 - Worked on IC design using TSMC 180nm and GPDK 90nm technologies.
 - Developed a Wishbone interface and validated a high-speed BiFET Op-Amp in Verilog.
- **Walton Hi-Tech Industries Limited** Dhaka, Bangladesh
Industrial Trainee(Nov 2022)
 - Observed and analyzed PCB manufacturing and EV assembly workflows.
 - Gained exposure to industrial electronics production workflows.

Standardized Test Score

- **IELTS Overall- 7.5/9** (Listening-8.5 , Reading-7.5 , Writing-6.5, Speaking-6.5) (Old score)

Undergraduate Projects

- **Traffic speed estimation system using Yolo-v2 network:** Gained hands-on experience generating large datasets and training a machine learning algorithm in MATLAB to detect vehicles and estimate their speed.
- **Development of robotic arm with 6 degrees of freedom:** Designed and simulated a robotic arm in SOLIDWORKS, built the model using 3D printing, and implemented PID control and inverse kinematics for precise operations.
- **Automatic drone delivery system using GPS:** Designed the drone body in SOLIDWORKS, used Arduino as the flight controller and PID for balancing, integrated GPS for self-navigation, and designed a payload deployment system for delivery.
- **Implementation of OMR (Optical Mark Recognition) machine using digital logic circuit:** Designed and simulated the OMR system, finally implemented it using NAND RAM, MUX, Comparator, and LDR, displayed the counted results.
- **Designing and building a 3D printing machine:** Used SOLIDWORKS for mechanical design, integrated an Arduino as the main controller and custom Marlin firmware for precise control and automation.
- **Real-time environmental monitoring system:** Developed a self-sustaining environmental monitoring system using Autodesk Eagle, Arduino, solar panels, and GSM network to collect data on humidity, temperature, and CO_2 concentration.
- **Detection of particle concentration using light scattering mechanism:** Used a laser as a light source in a dark chamber and a camera to detect scattering, utilized MATLAB for monitoring and analyzing particle concentration.
- **Design and implementation of a RISC-V processor with Wishbone interface:** Gained hands-on experience designing processors using Verilog in CADENCE, testing on FPGA, and implementing it using TSMC 180nm technology.
- **Design and implementation of laboratory variable DC power supply:** Performed comprehensive MATLAB analysis on design efficiency. Implemented both current and voltage control using an AC input and transformer for precise regulation.
- **Designing a regulated boost converter using TSMC's 180nm technology:** Utilized CADENCE tools for designing and optimizing for precise voltage control and energy efficiency through advanced circuit design and rigorous testing processes.

Skills Summary

- **Simulation & Design Software:** MATLAB Simulink, Cadence Virtuoso, PSPICE, PSAF, Icarus Verilog, GTKWave
- **Embedded Systems & MCU:** STM32 (HAL & LL), ESP32, AVR, Arduino, ARM Cortex-M, µVision Keil
- **Robotics & Middleware:** ROS (Robot Operating System)
- **Flight Control Systems:** ArduPilot, iNav, Betaflight
- **PCB & Circuit Design:** Autodesk Eagle, Altium Designer, KiCad, Proteus
- **CAD & Mechanical Design:** SolidWorks
- **3D Printing Platforms:** Marlin Firmware, Ultimaker Cura, Creality CR-6 SE
- **CNC Machining Software:** FlatCAM, Universal G-code Sender, GRBL
- **Programming Languages:** Python, C, Embedded C
- **Documentation:** Microsoft Office Suite, L^AT_EX
- **Soft Skills:** Leadership, Project Management, Time Management, Multitasking, Technical Reading

Honors and Awards

- University Merit Scholarship, BUET (three times - January 2019, January 2020, January 2022)
-Awarded to the students with outstanding academic record (GPA:3.90+) in a semester
- Dean's List Award, BUET (four times - In all academic years)
-Offered to the students obtaining an average GPA of 3.75+ in two consecutive semester of an academic year
- University Stipend (three times - January 2018, July 2018, July 2021)
- Higher Secondary Certificate (HSC-2017) Board Scholarship
- Runners-up, Bangabandhu Creative Talent Hunt. 2014, Pabna Division

Extra-curricular Activities

- Affiliate Member, BUET Robotics Society (BRS). (Jan 2022 to Mar 2023)
- Affiliate Member, Team Interplanetar - BUET Mars Rover Team. (Jun 2022 to Mar 2023)

References

Dr. Mahbub Alam, Associate Professor
Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
BUET, Dhaka-1000, Bangladesh.
Email: mahbubalam@eee.buet.ac.bd

Dr. Md. Ziaur Rahman Khan, Professor
Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
BUET, Dhaka-1000, Bangladesh.
Email: zrkhan@eee.buet.ac.bd